A MULTI-RATE AND AUTO-ADJUSTABLE WAVELET DECOMPOSITION FRAMEWORK FOR PATHOLOGICAL HAND TREMOR EXTRACTION

Soroosh Shahtalebi†, Arash Mohammadi†, Seyed Farokh Atashzar‡, and Rajni V. Patel‡

†Concordia Institute for Information Systems Engineering, Concordia University, Montreal, QC, Canada
‡ Electrical and Computer Engineering, Western University, London, ON, Canada

ABSTRACT

Estimation/extraction of pathological tremor signal is of paramount importance in different rehabilitation and clinical applications as it assists the physicians to evaluate the treatment response and investigate undiscovered aspects of tremor disorder. This paper is particularly motivated by the crucial need for having real-time and accurate estimate of the tremor signal in robotic rehabilitation systems to filter out (attenuate) the tremor from the voluntary movement of the patient. We propose a novel, multi-rate and auto-adjustable wavelet decomposition framework, referred to as the MAWD, for pathological hand tremor extraction. The proposed framework consists of two schemes running in parallel, i.e., hyper-parameter adjustment (HPA) scheme and real-time tremor prediction (RTP) scheme. More specifically the aforementioned schemes adaptively decompose the measurement signal consisting of both tremor and voluntary motion into different levels of approximation and use polynomial extrapolation to predict the value of the desired signal for the next time instant. The performance of the proposed framework is evaluated based on real pathological tremor data. The results indicate that the MAWD provides improved accuracy in comparison to the recently developed state-of-the-art pathological tremor estimation methodology.

***Index Terms*: Wavelet Transforms, Rehabilitation Systems, Pathological Tremor, Predictive Estimation.**

1. INTRODUCTION

Tremor [1] is described as an involuntary, rather rhythmic, and slightly sinusoidal movement which affects the coordination, targeting, and speed of motion [2–4]. Although tremor is not an intrinsically harmful disorder, it significantly reduces the quality of life as it reduces the patient's ability to perform the activities of daily living (ADL) [5]. This makes tremor estimation and extraction of paramount importance in various application domains ranging from assisted robotic surgery to assistive, wearable, and rehabilitative devices [9–13]. Typically, estimation/extraction of tremor is performed based on movement recordings obtained from multitude of sensor technologies ranging from accelerometer sensors to optical sensors [6], and is done either in an offline or online fashion. Offline methodologies provide more accurate results used for evaluation of treatment responses and are useful for discovering new properties of this disorder [7, 8]. On the other hand, online methods provide critical information about the voluntary and tremor signals and serve as the main building block of human-in-the-loop cyber physical systems (CPSs) for robotic rehabilitation and assistive systems.

The common problem among different tremor estimation and extraction techniques is that the recorded measurements contains both the voluntary movement and the tremor. This results in requiring online methodologies for separating the two components in real time and with high accuracy. Among different solutions [14–20] designed to address the need for online tremor estimation and extraction, adaptive filters developed based on Fourier linear combiners (FLC) and wavelet transforms have shown superior performance. The benefit of the FLC-based techniques is attributed to the fact that they impose minimum latency on the system making them appropriate for real-time interaction with robotic and rehabilitation devices. Several extension to the FLC technique are proposed in the literature including band-limited multiple FLC (BMFLC) [21, 22] and enhanced-BMFLC (E-BMFLC) [23]. Alternatively and due to the non-stationary nature of the tremor signals, there has been a surge of interest in deployment of wavelet transforms for tremor extraction [24–26] thanks to their multi-resolution behavior. Wavelet decomposition-based approaches provide spectral-temporal representation of the signal.

To the best of our knowledge, however, wavelet-based algorithms developed in the literature to analyze tremor signals only use wavelet coefficients as feature vectors to identify and discriminate tremor. The paper proposes an alternative and novel, multi-rate and auto-adjustable wavelet decomposition framework, referred to as the MAWD, for extracting pathological hand tremor signals. Intuitively speaking, recorded signals from hand of patients with tremor are inherently multi-scale due to contributions from events occurring with different localizations in time and frequency. The proposed MAWD framework decomposes the recorded signal into several spectra-temporal components deploying wavelet transforms and then predicts the tremor by means of a polynomial extrapolation. This MAWD framework consists of an auto-adjusting module (referred to as hyper-parameter adjustment (HPA) scheme) which runs in parallel to that of the main module (referred to as real-time tremor prediction (RTP) scheme). The HPA scheme is proposed to optimize and update the hyper-parameters of the RTP scheme. The proposed algorithm runs in real-time and predicts the tremor signal for one sample ahead, which is suitable for implementation on hardware and real world applications.

The rest of the paper is organized as follows. Section 2 formulates the problem and briefly reviews the required mathematical background. The proposed MAWD framework is presented in Section 3. Section 4 presents the experimental results for the evaluation of the proposed framework based on real pathological tremor data. Finally, Section 5 concludes the paper.

2. PROBLEM FORMULATION

We consider the following model for the hand movement of a patient with pathological tremor

$$x(t) = x_{p-v}(t) + x_{p-i}(t), \tag{1}$$

where $x(t)$ is the composite hand motion signal; $x_{p-v}(t)$ and $x_{p-i}(t)$ are, the voluntary and involuntary components of the motion, respectively, and; t is the time index. The recorded signal $x(t)$, therefore, conveys information about the patient's desired movement and the tremor that affects the desired movement. The goal of a tremor extraction approach is to estimate the involuntary component $x_{p-i}(t)$ in real-time. In the proposed MAWD framework, we deploy wavelet transforms to obtain multi-scale decomposition of the raw data. Wavelet transform decomposes a signal into its frequency components and provides scale-specific resolution components. It deploys a weighted summation of wavelet functions, referred to as the *Mother Wavelet*, to transform the signal into its temporal-spectral components. This is similar in nature to the Fourier transform used in the FLC-based tremor extraction methodologies, but unlike sines and cosines in Fourier transform, the mother wavelet and the ones generated from it are localized in space, therefore, providing a mechanism to approximate $x(t)$ by a series of scaled and translated versions of these localized functions. In other words, the wavelet transform decomposes the signal into different scales/resolutions with lower scales providing more details of high frequency components while the higher scales provide overall features associated with low frequency components. Before presenting the proposed MAWD framework, below we briefly present the wavelet decomposition technique.

Mathematically, a mother wavelet is a square integrable function $\psi(t)$, which satisfies the *admissibility condition* [27], i.e.,

$$0 < c_\psi = \int_{-\infty}^{\infty} \frac{|\hat{\psi}(\omega)|}{|\omega|} d\omega < \infty, \tag{2}$$

where $\hat{\psi}(\omega)$ is the Fourier transform of $\psi(t)$. Typically a regularity condition is preferable requiring $\psi(t)$ to be a fast decaying function or a nonzero function only on a finite interval [27]. To decompose a given function, wavelet transforms use a family of wavelet functions $\psi_{s,\tau}(t)$ obtained from the mother wavelet $\psi(t)$ by dilations and translations as follows

$$\psi_{s,\tau}(t) = \frac{1}{\sqrt{s}} \psi\left(\frac{t-\tau}{s}\right), \quad s, \tau \in \mathbb{R}, s \neq 0, \tag{3}$$

where s is the scale parameter, and τ is the location parameter. The continuous transform of a function $x(t)$ is then given by

$$T_x(s,\tau) = \int_{-\infty}^{\infty} x(t) \psi^*_{s,\tau}(t) dt, \tag{4}$$

where superscript * denotes the complex conjugate. The admissibility condition of $\psi(t)$ ensures that $x(t)$ can be recovered from $\psi_{s,\tau}(t)$ by the inverse transform [27] as follows

$$x(t) = \frac{1}{c_\psi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} T_x(s,\tau) \psi_{s,\tau}(t) \frac{d\tau ds}{s^2}, \tag{5}$$

which is referred to as the *wavelet decomposition* of signal $x(t)$. Variation of the scale and location parameters over a continuum of values in the continuous wavelet decomposition, however, leads to undesirable redundancy in the calculation of the wavelet coefficients. In practical applications, therefore, it is more convenient to sample the parameters s and τ on a discrete set of values in the scale-time plane. This leads to the *discrete wavelets* defined on suitably chosen grid points on the $s - \tau$ plane as follows

$$\psi_{j,k}(t) = s_0^{-j/2} \psi(s_0^{-j} t - k\tau_0), \quad j, k \in \mathbb{Z}, \tag{6}$$

where $s_0 > 1$ and τ_0 are fixed dilation and translation factors [27]. The so called *dyadic sampling* corresponds to the scenario when $s_0 = 2$ and $\tau_0 = 1$. The discrete wavelet transform (DWT) is then defined as follows

$$T_x(i,j) = \int_{-\infty}^{\infty} x(t) \psi^*_{j,k}(t) dt. \tag{7}$$

If the set of wavelets $\psi_{j,k}(t)$ form an orthogonal basis, the above transform can be inverted leading to the discrete wavelet decomposition of $x(t)$ given by

$$x(t) = \frac{1}{c_\psi} \sum_{j,k \in Z} T_x(j,k) \psi_{j,k}(t). \tag{8}$$

The DWT is especially suited for time series which are sampled at equal intervals of time. Orthonormal dyadic discrete wavelets are associated with scaling functions and their dilation equations. The scaling function associated with the smoothing of the signal has the same form as the wavelet, i.e.,

$$\phi_{j,k}(t) = s_0^{-j/2} \phi(s_0^{-j} t - k\tau_0), \quad j, k \in \mathbb{Z}, \tag{9}$$

with the property that $\int_{-\infty}^{\infty} \phi_{0,0}(t) = 1$, where $\phi_{0,0} = \phi$ is sometimes referred to as the father scaling function or father wavelet. The scaling function is orthogonal to translations of itself, but not to dilation of itself. The scaling function can be convolved with the signal to produce *approximation coefficients* as follows

$$S_{j,k} = \int_{-\infty}^{\infty} x(t) \phi_{j,k} dt. \tag{10}$$

The approximation coefficients at a specific scale j are collectively known as the discrete approximation of the signal at that scale. A continuous approximation of the signal at scale j can be generated by summing a sequence of scaling functions at this scale factored by the approximation coefficients as follows

$$x_j(t) = \sum_{k=-\infty}^{\infty} S_{j,k} \phi_{j,k}(t), \tag{11}$$

where $x_j(t)$ is a smooth, scaling-function-dependent version of the signal $x(t)$ at scale index j. This continuous approximation approaches $x(t)$ at small scales, i.e., as $j \to \infty$ [28]. A signal $x(t)$ can then be represented using a combined series expansion using both the approximated coefficients and the wavelet (detail) coefficients as

$$x(t) = \sum_{k=-\infty}^{\infty} S_{j_0,k} \phi_{j_0,k}(t) + \sum_{j=-\infty}^{m_0} \sum_{k=-\infty}^{\infty} T_{j,k} \psi_{j,k}(t). \tag{12}$$

This completes a brief introduction to the wavelet decomposition used in the proposed MAWD.

3. THE MAWD FRAMEWORK

The aim of the proposed MAWD framework is to extract the hand tremor signal from the raw measurements consisting of both voluntary and involuntary motions. The extracted signal is then utilized as the error signal for the feedback loop in a rehabilitation robotic system. The proposed MAWD simultaneously predicts the tremor in the next time instance and is capable of adaptively fine tuning its hyper-parameters, as the algorithm continues to operate. The MAWD is developed based on the following two multi-rate filtering schemes which run in parallel and cooperatively provide a self adjusting tremor extraction engine:

Algorithm 1 THE MAWD FRAMEWORK

Input: Raw measurement data recorded from patient movement; $x(t)$.

Output: Extracted tremor signal; $x_{p-i}(t)$.

S1. Collect L number of data samples as an input to the HPA to adjust the hyper-parameters for the RTP.

S2. Initialize first sample of tremor signal ($x_{p-i}(t_0)$) with value of the raw signal at that time instance ($x(t_0)$).

1: **for** L $\leq$ i $\leq$ End of signal x(t) **do**
2: **if** mod(i,L)== 0 **then**
3: Run the HPA for the last L samples to update the hyper-parameters and extract j^*.
4: **else**
5: Run the HPA on the last 2^M samples of raw signal.
6: Decompose the raw signal in M levels and derive the coefficients ($S_{j,k}$).
7: Reconstruct the signal with the approximation coefficients of $j^* - th$ level ($\hat{x}_{j^*}(t)$).
8: Extrapolate the reconstructed signal for the next time instance ($\hat{x}_{j^*}(t+1)$).
9: In next time instance, the difference between $x(t+1)$ and $\hat{x}_{j^*}(t+1)$, is tremor ($x_{p-i}(t+1)$).
10: **end if**
11: **end for**

1. *Hyper-Parameter Adjustment (HPA) Scheme*: The HPA filtering scheme performs post-processing on the previous time samples to derive the best hyper-parameters, therefore, enhancing and improving the prediction accuracy of tremor over time.
2. *Real-time Tremor Prediction (RTP) Scheme*: The RTP filtering scheme processes the raw sensory data in real-time and predicts the value of the tremor for the next time sample.

In the following subsections, we present each scheme in details.

3.1. Hyper-Parameter Adjustment (HPA) Scheme

The HPA filtering scheme performs post-processing on the previous samples of the raw signal by means of an off-line processing technique. Based on the result of the HPA scheme, a selection parameter is suggested to be used in the RTP filtering scheme. The HPA scheme is developed based on the available information in the literature regarding the characteristics of the tremor and its frequency contents. Previous studies suggest that the tremor occurs in a certain frequency band which (most of the time) is distinguishable from the frequency band associated with the desired movement. Therefore, by applying a sharp bandpass filtering to the signal obtained from the previous samples, we establish a ground truth representing the precisely extracted tremor based on previous samples of the signal. On the other hand, wavelet approximations of the signal at all levels of scaling are computed to be compared with the ground truth. Because of the time varying properties of the tremor signal, we analyze the past samples of the raw signal up to a certain amount of delay resulting in a sliding window of length L which selects last L samples of the raw signal to be fed into the HPA scheme. Mean square error (MSE), as a measure of similarity, is deployed to investigate the similarity of the ground truth to any of the computed approximations. The optimum level of approximation is denoted by j^* which is the level of approximation which yields the minimum error and is reported to the RTP scheme described next.

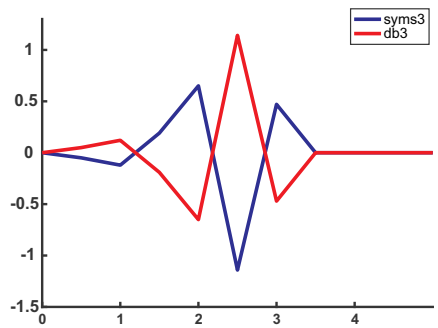

Fig. 1. The shape of *db3* and *syms3* mother wavelet functions.

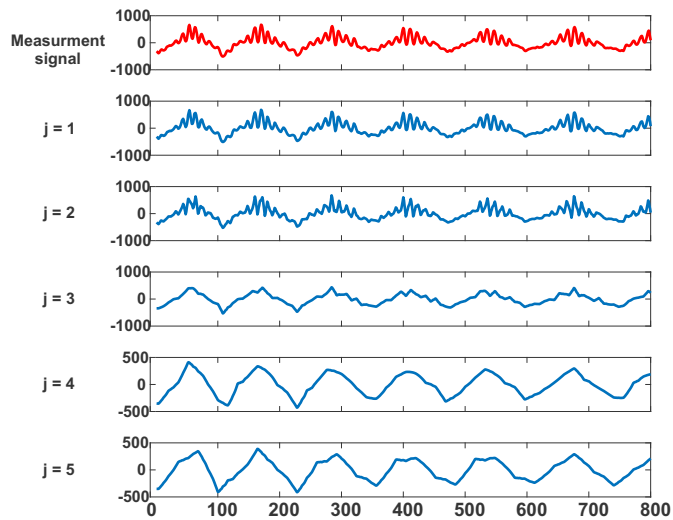

Fig. 2. Approximations of the measurement signal into 5 levels of decomposition using *db3* mother wavelet.

3.2. Real-time Tremor Prediction (RTP) Scheme

The RTP filtering scheme receives its hyper-parameter from the HPA scheme and performs the processing of the measurement signal in a real-time manner. If we decompose a signal in M levels, for ($1 \leq j \leq M$), at least 2^M samples of the signal are needed to perform the wavelet transformation. In the RTP scheme, the wavelet approximations of the ongoing signal for 2^M samples are computed and the signal is reconstructed based on the coefficients of the $j^* - th$ level of approximation provided from the HPA scheme. Practically, we believe that the $j^* - th$ level of approximation is the best representative of the desired signal, so if we subtract the reconstructed signal from the raw signal, we can extract the tremor. Then based on the reconstructed signal, an extrapolation is made to predict the value of the desired signal for the next time instance. Hence, the value of the signal in next time instance subtracted by the prediction in the previous run is considered as the value of the tremor.

Let us further elaborate on the overall procedure of the proposed MAWD framework for extracting the tremor signal. Denoting the raw measurement signal with $x(t)$, first, by analyzing $x(t)$ for the last L samples and performing the HPA scheme, we derive the parameter j^*. Next in the RTP scheme, by analyzing $x(t)$ for the last 2^M samples and reconstructing the signal with the approximation coefficients of the $j^* - th$ level of approximation, we derive

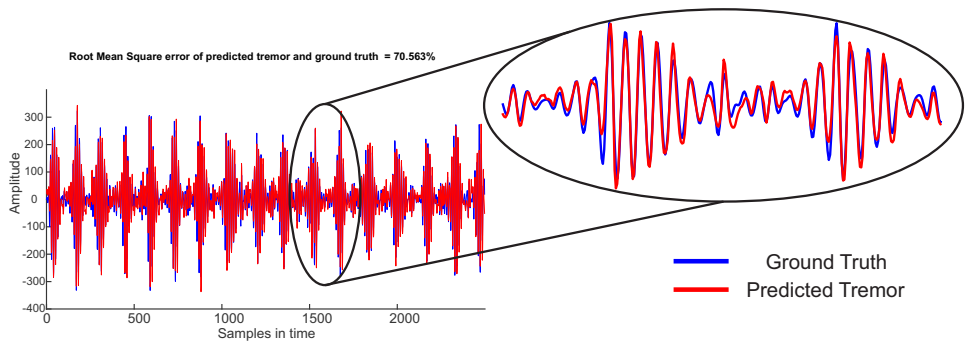

Fig. 3. Predicted tremor signal and its associated ground truth.

$\hat{x}_{j^*}(t)$. Then, by means of an extrapolation on $\hat{x}_{j^*}(t)$, we predict the value of approximation signal for the next time instance, which is $\hat{x}_{j^*}(t+1)$. In next time instance which is $t+1$, by subtracting $\hat{x}_{j^*}(t+1)$ from $x(t+1)$, we have $x_{p-i}(t+1)$, which is prediction of the tremor signal in time instant $t+1$. We initialize $x_{p-i}(t_0)$ by the value of the raw measurement signal at $(x(t_0))$. Algorithm 1 outlines the overall procedure of the proposed MAWD framework.

4. EXPERIMENTAL SETUP

In this section we evaluate the performance of the proposed MAWD framework based on a set of real pathological tremor data. The dataset used to evaluate the proposed algorithm is provided by Motus Bioengineering Incorporation [29] which includes real measured signals from patients with pathological hand tremor. The dataset consists of recordings from both left and right hands. The dataset includes bi-axial signals of 100 Hz corresponding to hand tremor collected via a Motus Gyroscope which records angular velocities where the unit of the recordings is "degree/second".

As stated previously, the frequency band corresponding to the tremor signal is typically distinguishable from the one associated with the desired movement. By analyzing the frequency contents of the recorded signals from various patients, it is observed that the frequency content of the tremor is mostly located between 6-14 Hz. In the HPA scheme, therefore, we deploy a sharp bandpass filter to extract the frequency contents between 6-14 Hz which contains most of the frequency contents of the tremor signal. On the other hand, we have evaluated effects of choosing different wavelet mother functions on the overall performance of the proposed MAWD. It is observed that Daubechies and Symlets mother functions of order 3 provide the highest performance in extraction of the tremor. Therefore, during all the experiments we utilize the Daubechies-3 also known as *db3*. The shape of the two aforementioned mother wavelet functions are depicted in Fig. 1. We would like to point out that the mother function does not change during the experiments and is fixed for both HPA and RTP schemes.

In our experiments, it is observed that incorporation of only five levels of approximation is enough to yield the required performance. Besides, the output of the HPA scheme commonly varies between $j = 2, 3$ or 4. Moreover, it is worth mentioning that for the prediction step, we use a polynomial extrapolation of order 2. Fig. 2 illustrates approximations of the measurement signal into five levels of decomposition.The signal used to generate Fig. 2 is recorded from a patient with action tremor on his/her left hand where the patient with action tremor is asked to perform voluntary Pronation/Supination in his/her left hand. When the proposed algorithm starts processing of the raw measurements, at first L number of samples from the signal at hand is provided to the HPA scheme. In our experiments, the window size, L, is considered to be equal to 500 samples. In other words and based on the sampling frequency of $100Hz$, the initial window size used in HPA is five seconds of signal. Selection of $L = 500$ enables the HPA scheme to more accurately analyze the signal and compute the j^* parameter in an optimized fashion. The proposed framework is evaluated on the aforementioned dataset and the results are depicted in Fig. 3. The results show that the mean square error between the ground truth signal for the tremor and extracted tremor, is 70.563%. Please note that the ground truth signal in this part is extracted from the whole measurement signal and is not derived from a windowed part of signal. In comparison to the performance obtained from the Least mean square (LMS)-based E-BMFLC [23], the proposed MAWD framework provides about 5% improvement. As a final comparison measure, we compute the normalized root mean-square error (NRMSE), defined as $\text{NRMSE} = \sqrt{\sum_{k=1}^{n}(\hat{x}_{p-i}(k) - x_{p-i}(k))^2}/(x_{\max} - x_{\min})$, where $x_{p-i}(k)$ is the input tremor, $\hat{x}_{p-i}(t)$ is the estimated tremor, and n is the number of samples. Terms $x_{\max}$ and $x_{\min}$, respectively, denote the minimum and maximum value of the observed signal. The NRMSE result based on the proposed MAWD is 0.0405 which illustrates a notable improvement in comparison to the NRMSE of 0.0551 which is obtained based on E-BMFLC [23] the state-of-the-art tremor estimation algorithm.

5. CONCLUSION

In this paper, we proposed a new technique to extract the tremor signal out of the measurement data which is based on wavelet transforms. Measurement data is a combination of the signals related to patient's desired movement and patient's tremor. The algorithm is evaluated on a dataset and the MSE measure for accuracy of prediction is 70.563%. We believe that the accuracy of the proposed method could be further increased by optimizing some parts of the algorithm. As future directions of this work, we can think of replacing the polynomial extrapolation step by a better and optimized predictor. Moreover, the selection of the frequency band of $6 - 14\ Hz$ could be done by an adaptive algorithm, since we believe that the aforementioned frequency band varies from subject to subject. The other development to this algorithm could be an adaptive technique to select the mother wavelet function.

6. REFERENCES

[1] D. Tarsy, "Handbook of essential tremor and other tremor disorders," *Neurology*, vol. 70, no. 4, pp. 326-326, 2008.

[2] J. Shahed and J. Jankovic, "Exploring the Relationship Between Essential Tremor and Parkinson's Disease," *Parkinsonism & Related Disorders*, vol. 13, no. 2, pp. 67-76, 2007.

[3] E.D. Louis and J.P.G. Vonsattel, "The Emerging Neuropathology of Essential Tremor," *Movement Disorders*, vol. 23, no. 2, pp. 174-182, 2008.

[4] K. Wirdefeldt, H.-O. Adami, P. Cole, D. Trichopoulos, and J. Mandel, "Epidemiology and Etiology of Parkinson's Disease: A Review of the Evidence," *European Journal of Epidemiology*, vol. 26, no. 1, 2011.

[5] R. Elble and W. Koller, "Tremor, Johns Hopkins Series in Contemporary Medicine and Public Health," 1990.

[6] J.P. Giuffrida, D.E. Riley, B.N. Maddux, and D. A. Heldman, "Clinically Deployable Kinesia Technology for Automated Tremor Assessment," *Movement Disorders*, vol. 24, no. 5, pp. 723-730, 2009.

[7] J. Timmer, "Modeling Noisy Time Series: Physiological Tremor," *International Journal of Bifurcation and Chaos*, vol. 8, no. 07, pp. 1505-1516, 1998.

[8] J. Spyers-Ashby, P. Bain, and S. Roberts, "A Comparison of Fast Fourier Transform (FFT) and Autoregressive (AR) Spectral Estimation Techniques for the Analysis of Tremor Data," *Journal of Neuroscience Methods*, vol. 83, no. 1, pp. 35-43, 1998.

[9] E. Rocon, M. Manto, J. Pons, S. Camut, and J. M. Belda, "Mechanical Suppression of Essential Tremor," *The Cerebellum*, vol. 6, no. 1, pp. 73-78, 2007.

[10] E. Rocon, J. Belda-Lois, A. Ruiz, M. Manto, J. C. Moreno, and J. Pons, "Design and Validation of a Rehabilitation Robotic Exoskeleton for Tremor Assessment and Suppression," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 15, no. 3, pp. 367-378, 2007.

[11] L.P. Maneski, N. Jorgovanović, V. Ilić, S. Došen, T. Keller, M. B. Popović, and D. B. Popović, "Electrical Stimulation for the Suppression of Pathological Tremor," *Medical & Biological Engineering & Computing*, vol. 49, no. 10, 2011.

[12] R.J. OConnor and M. U. Kini, "Non-Pharmacological and Non-Surgical Interventions for Tremor: A Systematic Review," *Parkinsonism & Related Disorders*, vol. 17, no. 7, pp. 509-515, 2011.

[13] K. Kiguchi and Y. Hayashi, "An EMG-based Control for an Upper-Limb Power-Assist Exoskeleton Robot," *IEEE Transactions on Systems, Man, and Cybernetics, Part B (Cybernetics)*, vol. 42, no. 4, pp. 1064-1071, 2012.

[14] A. Kucukelbir, A. Kushki and K.N. Plataniotis, "A New Stochastic Estimator for Tremor Frequency Tracking," *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 421-424, 2009.

[15] S. Dosen, S. Muceli, J.L. Dideriksen, J.P. Romero and E. Rocon, J. Pons and D. Farina, "Online Tremor Suppression Using Electromyography and Low-Level Electrical Stimulation," in *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 3, pp. 385-395, 2015.

[16] K.C. Veluvolu and W.T. Ang, "Estimation and Filtering of Physiological Tremor for Real-Time Compensation in Surgical Robotics Applications," *Int. J. Medical Robotics and Computer Assisted Surgery*, vol. 6, pp. 334-342, 2010.

[17] K.C. Veluvolu, W.T. Latt, and W.T. Ang, "Double Adaptive Fourier Linear Combiner for Real-TIme Estimation/Filtering of Tremor," *J. Biomed. Signal Process. & Control*, vol. 5, pp. 33-44, 2010.

[18] K.C. Veluvolu and W.T. Ang, "Estimation of Physiological Tremor from Accelerometers for Real-Time Applications," *Sensors*, vol. 11, no. 3, pp. 3020-3036, 2011.

[19] K.C. Veluvolu, S. Tatinati, S. M. Hong and W. T. Ang, "Multistep Prediction of Physiological Tremor for Surgical Robotics Applications," in *IEEE Trans. Biomed. Eng.*, vol. 60, no. 11, pp. 3074–3082, 2013.

[20] S. Wang, Y. Gao, J. Zhao, and H. Cai, "Adaptive Sliding Bandlimited Multiple Fourier Linear Combiner for Estimation of Pathological Tremor," *Biomed. Sig. Process. & Control*, vol. 10, pp. 260-274, 2014.

[21] K.C. Veluvolu, U.X. Tan, W.T. Latt, C. Shee, and W.T. Ang, "Bandlimited Multiple Fourier Linear Combiner for Real-Time Tremor Compensation," *IEEE International Conference of the Engineering in Medicine and Biology Society*, pp. 2847-2850, 2007

[22] K. Veluvolu and W. Ang, "Estimation and Filtering of Physiological Tremor for Real-Time Compensation in Surgical Robotics Applications," *The International Journal of Medical Robotics & Computer Assisted Surgery*, vol. 6, no. 3, pp. 334-342, 2010.

[23] S.F. Atashzar, M. Shahbazi, O. Samotus, M. Tavakoli, M.S. Jog, and R.V. Patel, "Characterization of Upper-Limb Pathological Tremors: Application to Design of An Augmented Haptic Rehabilitation System," *IEEE Journal of Selected Topics in Signal Processing*, vol. 10, no. 5, pp. 888-903, 2016.

[24] A. Hossen, M. Muthuraman, J. Raethjen, G. Deuschl, and U. Heute, "Discrimination of Parkinsonian Tremor from Essential Tremor by Implementation of a Wavelet-based Soft-decision Technique on EMG and Accelerometer Signals," *Biomedical Signal Processing and Control*, vol. 5, no. 3, pp. 181-188, 2010.

[25] O. Geman and C. Zamfir, "Using Wavelet for Early Detection of Pathological Tremor," *European Signal Processing Conference (EUSIPCO)*, pp. 1723-1727, 2012.

[26] L. Popovic and M.B. Popovic, "Extraction of tremor for control of neural prostheses: Comparison of discrete wavelet transform and butterworth filter," *IEEE Symposium on Neural Network Applications in Electrical Engineering*, pp. 137-140, 2008.

[27] I. Daubechies, *Ten Lectures on Wavelets, SIAM*, 1992.

[28] P.S. Addison, "Wavelet Transforms and the ECG: A Review," *Physiological Measurement*, vol. 26, no. 5, 2005.

[29] G.N. Reeke, "Modeling in the Neurosciences: From Biological Systems to Neuromimetic Robotics," *CRC Press*, 2005'.